

The function of directing the movement of goods through the entire manufacturing cycle from the requisitioning of raw materials to the inventory of finished goods in an orderly manner to meet the objectives of maximum customer service with minimum investment and efficient (low cost) plant operation is termed as *inventory control*.

8.2.1. Classification of Inventories

Inventories may be classified as those which play direct role during manufacture or which can be identified on the product and the second one are those which are required for manufacturing but not as a part of production or cannot be identified on the product. The first type is labeled as *direct inventories* and the second are labeled as *indirect inventories*.

Further classification of direct and indirect inventories is as follows:

(A) Direct inventories

- (i) **Raw material inventories:** The inventory of raw materials is the materials used in the manufacture of product and can be identified on the product. In inventory control manager can concentrate on the
 - (a) Bulk purchase of materials to save the investment,
 - (b) To meet the changes in production rate,
 - (c) To plan for buffer stock or safety stock to serve against the delay in delivery of inventory against orders placed and also against seasonal fluctuations.
- (ii) **Work-in-process inventories or in process inventories:** These inventories are of semi-finished type, which are accumulated between operations or facilities. As far as possible, holding of materials between operations to be minimized if not avoided. This is because; as we process the materials the economic value (added labour cost) and use value are added to the raw material, which is drawn from stores. Hence if we hold these semi finished material for a long time the inventory carrying cost goes on increasing, which is not advisable in inventory control. These inventories serves the following purpose:
 - (a) Provide economical lot production,
 - (b) Cater to the variety of products,
 - (c) Replacement of wastages,
 - (d) To maintain uniform production even if sales varies.
- (iii) **Finished goods inventories:** After finishing the production process and packing, the finished products are stocked in stock room. These are known as finished goods inventory. These are maintained to:
 - (a) To ensure the adequate supply to the customers,
 - (b) To allow stabilization of the production level and
 - (c) To help sales promotion programme.
- (iv) **Spare parts inventories:** Any product sold to the customer, will be subjected to wear and tear due to usage and the customer has to replace the worn-out part. Hence the manufacturers always calculate the life of the various components of his product and try to supply the spare components to the market to help after sales service. The use of such spare parts inventory is:
 - (a) To provide after sales service to the customer,
 - (b) To utilize the product fully and economically by the customer.

- (iv) **Scrap or waste inventory:** While processing the materials, we may come across certain wastages and certain bad components (scrap), which are of no use. These may be used by some other industries as raw material. These are to be collected and kept in a place away from main stores and are disposed periodically by auctioning.

(B) Indirect Inventories

Inventories or materials like oils, grease, lubricants, cotton waste and such other materials are required during the production process. But we cannot identify them on the product. These are known as indirect inventories. In our discussion of inventories, in this chapter, we only discuss about the direct inventories.

Inventories may also be classified depending their nature of use. They are:

- (i) **Fluctuation Inventories:** These inventories are carried out to safeguard the fluctuation in demand, non-delivery of material in time due to extended lead-time. These are some times called as Safety stock or reserves. In real world inventory situations, the material may not be received in time as expected due to trouble in transport system or some times, the demand for a certain material may increase unexpectedly. To safeguard such situations, safety stocks are maintained. The level of this stock will fluctuate depending on the demand and lead-time etc.
- (ii) **Anticipation inventory:** When there is an indication that the demand for company's product is going to be increased in the coming season, a large stock of material is stored in anticipation. Some times in anticipation of raising prices, the material is stocked. Such inventories, which are stocked in anticipation of raising demand or raising rises, are known as anticipation inventories.
- (iii) **Lot size inventory or Cycle inventories :** This situation happens in batch production system. In this system products are produced in economic batch quantities. It some time happens that the materials are procured in quantities larger than the economic quantities to meet the fluctuation in demand. In such cases the excess materials are stocked, which are known as lot size or cycle inventories.
- (iv) **Transportation Inventories:** When an item is ordered and purchased they are to be received from the supplier, who is at a far of distance. The materials are shipped or loaded to a transport vehicle and it will be in the vehicle until it is delivered to the receiver. Similarly, when a finished product is sent to the customer by a transport vehicle it cannot be used by the purchaser until he receives it. Such inventories, which are in transit, are known as Transportation inventories.
- (v) **Decoupling inventories:** These inventories are stocked in the manufacturing plant as a precaution, in case the semi finished from one machine does not come to the next machine, this stock is used to continue a production. Such items are known as decoupling inventories.

8.3. COSTS ASSOCIATED WITH INVENTORY

While maintaining the inventories, we will come across certain costs associated with inventory, which are known as **economic parameters**. Most important of them are discussed below:

- (A) **Inventory Carrying Charges, or Inventory Carrying Cost or Holding Cost or Storage Cost (C_1) or ($i\%$):**

This cost arises due to holding of stock of material in stock. This cost includes the cost of maintaining the inventory and is proportional to the quantity of material held in stock and the time for which the material is maintained in stock. The components of inventory carrying cost are:

- (i) Rent for the building in which the stock is maintained if it is a rented building. In case it is own building, depreciation cost of the building is taken into consideration. Sometimes for own buildings, the nominal rent is calculated depending on the local rate of rent and is taken into consideration.
- (ii) It includes the cost of equipment if any and cost of racks and any special facilities used in the stores.
- (iii) Interest on the money locked in the form of inventory or on the money invested in purchasing the inventory.
- (iv) The cost of stationery used for maintaining the inventory.
- (v) The wages of personnel working in the stores.
- (vi) Cost of depreciation, insurance.
- (vii) Cost of deterioration due to evaporation, spoilage of material etc.
- (viii) Cost of obsolescence due to change in requirement of material or changed in process or change in design and item stored as a result of becomes old stock and become useless.
- (ix) Cost of theft and pilferage *i.e.* indenting for the material in excess of requirement.

This is generally represented by C_1 rupees per unit quantity per unit of time for production model. That is manufacturing of items model. For purchase models it is represented by $i\%$ of average inventory cost.

If we take practical situation into consideration, many a time we see that the inventory carrying cost (some of the components of the cost) cannot be taken proportional to the quantity of stock on hand. For example, take rent of the stores building. As and when the stock is consumed, it is very difficult to calculate proportion of rent in proportion to the stock in the stores as the rent will not vary day to day due to change in inventory level. Another logic is that the money invested in inventory may be invested in other business or may be deposited in the bank to earn interest. As the money is in the form of inventory, we cannot earn interest but loosing the expected interest on the money. This cost of money invested, is generally compared to the interest rate $i\%$ and is taken as the inventory carrying cost. Hence the value of ' i ' will be a fraction of a rupee and will be $0 < i < 1$. In many instances, the bank rate of interest is somewhere between 16 to 20 % and other components like salary, insurance, depreciation etc may work out to 3 to 5 %. Hence, the total of all components will be around 22 to 25 % and this is taken as the cost of inventory carrying cost and is expressed as $i\%$ of average inventory cost.

(B) Shortage cost or Stock - out - cost- (C_2)

Some times it so happens that the material may not be available when needed or when the demand arises. In such cases the production has to be stopped until the procurement of the material, which may lead to miss the delivery dates or delayed production. When the organization could not meet the delivery promises, it has to pay penalty to the customer. If the situation of stock out will occur very often, then the customer may not come to the organization to place orders, that is the organization is loosing the customers. In other words, the organization is loosing the goodwill of the customers. The cost of good will cannot be estimated. In some cases it will be very heavy to such extent that the

organization has to forego its business. Here to avoid the stock out situation, if the organization stocks more material, inventory carrying cost increases and to take care of inventory cost, if the organization purchase just sufficient or less quantity, then the stock out position may arise. Hence the inventory manager must have sound knowledge of various factors that are related to inventory carrying cost and stock out cost and estimate the quantity of material to be purchased or else he must have effective strategies to face grave situations. The cost is generally represented as so many rupees and is represented by C_2 .

(C) Set up cost or Ordering cost or Replenishment Cost (C_3)

For purchase models, the cost is termed as ordering cost or procurement cost and for manufacturing cost it is termed as set up cost and is represented by C_3 .

(i) **Set up cost:** The term set up cost is used for production or manufacturing models. Whenever a job is to be produced, the machine is to set to produce the job. That is the tool is to be set and the material is to be fixed in the jobholder. This consumes some time. During this time the machine will be idle and the labour is working. The cost of idle machine and cost of labour charges are to be added to the cost of production. If we produce only one job in one set up, the entire set up cost is to be charged to one job only. In case we produce ' n ' number of jobs in one set up, the set up cost is shared by ' n ' jobs. In case of certain machines like N.C machines, or Jig boarding machine, the set up time may be 15 to 20 hours. The idle cost of the machine and labour charges may work out to few thousands of rupees. Once the machine set up is over, the entire production can be completed in few hours. If we produce more number of products in one set up the set up cost is allocated to all the jobs equally. This reduces the production cost of the product. For example let us assume that the set up cost is Rs. 1000/-. If we produce 10 jobs in one set up, each job is charged with Rs. 100/- towards the set up cost. In case, if we produce 100 jobs, the set up cost per job will be Rs. 10/-. If we produce, 1000 jobs in one set up, the set up cost per job will be Re. 1/- only. This can be shown by means of a graph as shown in figure 8.1.

(ii) **Ordering Cost or Replenishment Cost :** The term Ordering cost or Replenishment cost is used in purchase models. Whenever any material is to be procured by an organization, it has to place an order with the supplier. The cost of stationary used for placing the order, the cost of salary of officials involved in preparing the order and the postal expenses and after placing the order enquiry charges all put together, is known as Ordering cost. In Small Scale Units, this may be around Rs. 25/- to Rs. 30/- per order. In Larger Scale Industries, it will be around Rs. 150 to Rs. 200 /- per order. In Government organizations, it may work out to Rs. 500/- and above per order. If the organization purchases more items per order, all the items share the ordering cost. Hence the materials manager must decide how much to purchase per order so as to keep the ordering cost per item at minimum. One point we have to remember here, to reduce the ordering cost per item, if we purchase more items, the inventory carrying cost increases. To keep inventory carrying cost under control, if we purchase less quantity, the ordering cost increase. Hence one must be careful enough to decide how much to purchase? The nature of ordering cost can also be shown by a graph as shown in figure 8.1. If the ordering cost is C_3 per order (can be equally applied to set up cost) and the quantity ordered / produced is ' q ' then the ordering cost or set up cost per unit will be C_3/q is inversely proportional to the quantity ordered, *i.e.* decreased with the increase in ' q ' as shown in the graph 8.1.

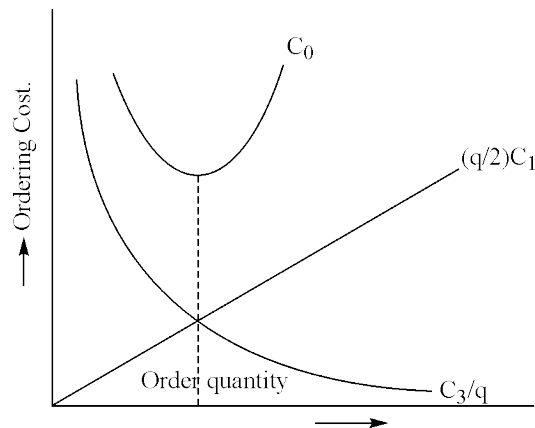


Figure 8.1 Nature of ordering cost.

- (iii) **Procurement Cost** : These costs are very much similar to the ordering cost / set up cost. This cost includes cost of inspection of materials, cost of returning the low quality materials, transportation cost from the source of material to the purchaser's site. This is proportional to the quantity of materials involved. This cost is generally represented by 'b' and is expressed as so many rupees per unit of material. For convenience, it always taken as a part of ordering cost and many a time it is included in the ordering cost / set up cost.

(D) Purchase price or direct production cost

This is the actual purchase price of the material or the direct production cost of the product. It is represented by 'p'. i.e. the cost of material is Rs. 'p' per unit. This may be constant or variable. Say for example the cost of an item is Rs. 10/- item if we purchase 1 to 10 units. In case we purchase more than 10 units, 10 percent discount is allowed. i.e. the cost of item will be Rs.9/- per unit. The purchase manager can take advantage of discount allowed by purchasing more. But this will increase the inventory carrying charges. As we are purchasing more per order, ordering cost is reduced and because of discount, material cost is reduced. Materials manager has to take into consideration these cost – quantity relationship and decide how much to purchase to keep the inventory cost at low level.

Points to be remembered

- (i) Inventory cost increases with the quantity purchased.
- (ii) If we purchase more items per order or produce more items per set up ordering cost or set up cost per item decreases, stock out situation reduces and inventory-carrying cost increases and if discount is allowed on quantity purchased the material cost also reduces.
- (iii) If we purchase less items per order or produce less items per set up ordering cost per item or set up cost per item increases, stock out position may increase which increases stock out costs, and inventory-carrying cost decreases. Quantity discounts may not be available.